An Extended Mechanism for Chemical Activation Systems

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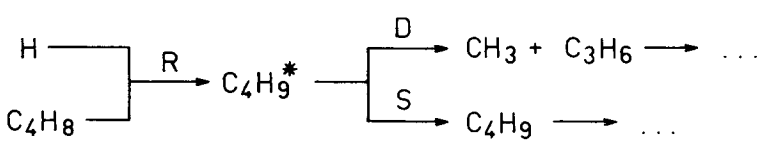
Abstract

The classical mechanism of chemically activated unimolecular reactions is extended to interpret experimental results in systems olefin/hydrogen atoms. In the case of a large excess of the latter, one has to take into consideration a second chemical activation of the primarily formed radicals by their recombination with H-atoms to yield chemically activated alkane molecules. The radicals as well as the alkane molecules may decompose, and a method based on the coupling of two steady-state master equations is developed to evaluate the contribution of either of the two reaction paths to the ratio decomposition to stabilization. The treatment proposed is demonstrated for the example *cis* but-2-ene/excess of thermal hydrogen atoms in the presence of molecular hydrogen as a bath gas at 298 K. It turns out that the step via activated butane is not negligible under these conditions.

Introduction

The classical treatment of chemically activated unimolecular reactions was developed in the 1960s on systems H-atom/olefin mainly by Rabinovitch and coworkers [1–3]. Experimental conditions were carefully chosen to have always a large excess of substrate over attacking H-atoms in order to suppress consecutive reactions caused by the latter. In this case, the main reaction of the hydrogen atoms is the addition to the olefin, and all other reactions are neglected. Scheme I demonstrates the reaction mechanism for the example of *cis* but-2-ene:



Scheme I

(asterisk denotes chemically activated species; R, D, and S are the rates of formation, decomposition, and collisional stabilization of the 2-butyl radical, respectively).

The variable, usually readily obtainable from experiments, is the ratio of decomposition to stabilization, D/S, which can be provided theoretically by solving the appropriate master equation [e.g., 4,5]. Recent works deal with the influence of the models of the probabilities for collisional energy transfer, with problems of the steady-state approximation, and with the relation between thermal and chemical activation [6–9]. The purpose of the present

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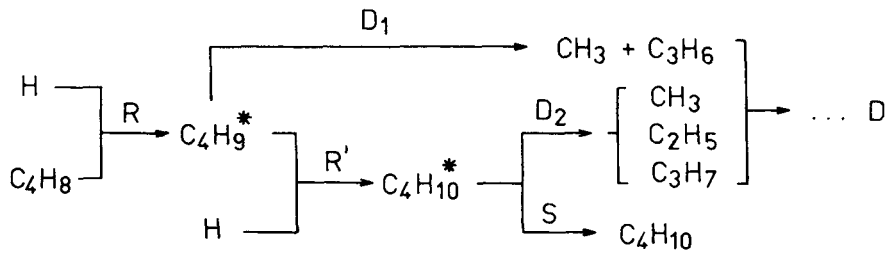
article is to describe the mechanistic extension and its consequences on the kinetic parameters if the requirement of excess of olefin is dropped. In this case, one has to allow for additional reactions of the hydrogen atoms with the chemically activated radicals and with reaction products. For mechanistic simplicity, we investigate the case opposite to the pioneering work, the case of high excess of hydrogen atoms over olefin in the presence of molecular hydrogen as a bath gas. The following consequences arise: The primarily formed radical may decompose or may be deactivated by collisions, mainly with the bath gas. The only chemical reaction in competition to the decomposition is the recombination with a hydrogen atom. In each case, this reaction leads to a chemically activated alkane molecule, the internal energy of which depends on the degree of collisional stabilization of the parent radical. This alkane molecule is now subjected to the competition between collisional stabilization and decomposition. Therefore, we have to extend the above Scheme I in the following way:



Scheme II

(with symbols as noted above, but in what follows the letter S in each case refers to collisional stabilization of the alkane). In our system at room temperature (298 K), the bimolecular reaction

$$2\text{-}C_4H_9 + H_2 \longrightarrow C_4H_{10} + H$$

should play a minor role, because, notwithstanding the vibrational excitation of 2-C_4H_9, the translational degrees of freedom, which mainly determine bimolecular reactions, are thermalized [see 10].

The selfcombination of the 2-butyl radicals is effectively suppressed by the excess of H-atoms, and no dimethylhexane was found experimentally.

Consecutive reactions of the decomposition products with hydrogen atoms, however, do not allow to assign the products to either of the two decomposition paths. Thus, the easily measurable ratio D/S represents the sum $(D_1 + D_2)/S$, and there arises the question for the ratio D_2/D_1.

To our knowledge, the first remark regarding this problem came from Hoyermann and coworkers [11], but until now the question has not been investigated in detail. In this work, we propose a method, based on master equation analysis, to evaluate the ratio D_2/D_1 from the ratio D/S.

Experimental

The measurements have been carried out at room temperature in an isothermal flow system with a preceding Wood-Bonhoeffer discharge. The experimental equipment is described elsewhere [12]. The partial pressures

at the inlet of the *cis* but-2-ene were: P_{H_2} = 133 Pa, P_H = 21.3 Pa, and $P_{C_4H_8}$ < 2 Pa. The ratio of the diameter of the flow tube to the mean free path is about 130. The analysis of the products by conventional gas chromatography provided a ratio $D/S = 2.08 \pm 0.04$ (one standard deviation). The ratio was determined by carbon balance, with methane, ethane, ethylene, propane, and propylene being taken into account for D, and *n*-butane represents stabilization S. More than 95% of the overall carbon were found in the above products [see also 30].

Under very similar conditions, but with a high excess of *cis* but-2-ene over hydrogen atoms in the presence of H_2 as bath gas (P_{H_2} = 145 Pa, $P_{C_4H_8}$ = 15 Pa), Rabinovitch and Diesen [1] found a value $D/S = 0.96$. We take this difference as an indication that under our conditions the second reaction path plays an important role.

Theory

Scheme II represents the starting point for the theoretical consideration. At first, the master equation for the 2-butyl radicals is formulated (we consider energy resolution as sufficient and neglect angular momentum dependence) [e.g., 4]:

$$\frac{dn_i}{dt} = Rf_i - \omega n_i + \omega_{nr} \sum_j P_{ij} n_j - k_i n_i . \tag{1}$$

Here n_i is the concentration of 2-butyl radicals with internal energy E_i, Rf_i is the rate of noncollisional input into, and k_i the rate constant for the unimolecular reaction out of this level.

In the case of more than one possible reaction, the sum of the specific rate constants of all parallel channels has to be taken (ethyl and methyl rupture in the case of butane, see below). In each case, however, we neglect the reverse process to the formation of the chemically activated species (C—H bond cleavage) because of its higher threshold energy compared with the reactions of interest (C—C bond cleavage). P_{ij} denotes the collisional transition probability for the transition $j \rightarrow i$.

One peculiarity in our system is the occurrence of reactive collisions between H and 2-butyl. This requires a distinction between the collisional frequencies ω and ω_{nr} in eq. (1): Every collision, reactive or nonreactive, removes a radical from level i (ω, second term) but only nonreactive collisions contribute to transitions $j \rightarrow i$ (ω_{nr}, third term).

Under steady-state conditions in the usual matrix formulation [3,4] eq. (1) can be rewritten:

$$R\mathbf{F} = (\omega\mathbf{I} - \omega_{nr}\mathbf{P} + \mathbf{K})\mathbf{N}^s = \mathbf{J}\mathbf{N}^s, \tag{2}$$

and the steady-state population N^s is obtained formally by

$$\mathbf{N}^s = R\mathbf{J}^{-1}\mathbf{F}. \tag{3}$$

The vector/matrix symbols in eq. (2) correspond to symbols in eq. (1), and $\mathbf{I}$ denotes the unit matrix.

Decomposition D_1 is defined by [3,4]:

$$D_1 \equiv \sum_i (\mathbf{K}\mathbf{N}^s)_i = R \sum_i (\mathbf{K}\mathbf{J}^{-1}\mathbf{F})_i , \tag{4}$$

and one obtains

$$\frac{D_1}{R} = \sum_i (\mathbf{K}\mathbf{J}^{-1}\mathbf{F})_i, \tag{5}$$

where $(\ )_i$ means the i-th diagonal element of the corresponding matrix. Equation (5) provides a means of calculating the ratio of decomposition to formation for any normalized input distribution $\mathbf{F}$.

Because of the high excess of atomic hydrogen, we suppose that all radicals that do not decompose, react with hydrogen atoms and hence do not contribute to stabilization S. Thus, we get according to Scheme II:

$$R = D_1 + R'. \tag{6}$$

Because R is an unknown but scalar quantity, it does not change the shape of the distribution in eq. (3), and a normalized steady-state distribution ($\hat{}$ stands for normalization) can be obtained by:

$$\hat{\mathbf{N}}^s = \frac{\mathbf{J}^{-1}\mathbf{F}}{\sum_i (\mathbf{J}^{-1}\mathbf{F})_i} \tag{7}$$

Obeying this distribution, 2-butyl radicals meet thermal hydrogen atoms to form n-butane. Following a similar approach by Knox [13, cf. also 5, page 212], we express the distribution f_i' (or $f'(E)$ on a continuous scale) of the butane molecules in the moment of their formation by the convolution of the steady-state distribution of 2-butyl radicals over the thermal distribution of the H-atoms and shift the resulting distribution by the bond dissociation energy D_0 of a secondary C—H bond in butane towards higher energies:

$$f'(E) = \int_0^{E-D_0} g_{\mathrm{H}}^{\mathrm{eq}}(\varepsilon)\hat{n}^s_{\mathrm{C_4H_9}}(E - D_0 - \varepsilon)\, d\varepsilon. \tag{8}$$

For $g_{\mathrm{H}}^{\mathrm{eq}}(\varepsilon)$, we take the positive part of a one-dimensional Maxwellian for hydrogen atoms, because only positive momenta contribute to the recombination, and the steady-state distribution $\hat{n}^s_{\mathrm{C_4H_9}}$ is given by eq. (7). Deriving eq. (8), we assume that the dependence of the cross-section for combination on the relative velocity of the reactants can be neglected because of the very narrow velocity distribution at room temperature. Furthermore, we assume that the cross-section is also independent of the vibrational excitation of the 2-butyl radical. We feel that a certain justification results from the fact that the excitation energy ($\lesssim$15 000 cm^{-1}, see $\hat{n}^s$ in Figs. 1 and 2) is considerably smaller than the bond dissociation energy of the C—H bond formed ($\approx$35 000 cm^{-1}).

Because energy randomization is fast compared with unimolecular dissociation, a C—C bond rupture is favored over a C—H bond rupture, whereas only the latter would decrease the cross-section of the recombination reaction. For bimolecular reactions with an activation barrier, no dependence of the reaction rate on vibrational excitation of one of the reactants was found [10 and literature cited therein]. In our practical calculations, the finite-difference analogon of eq. (8) is used (see the following section).

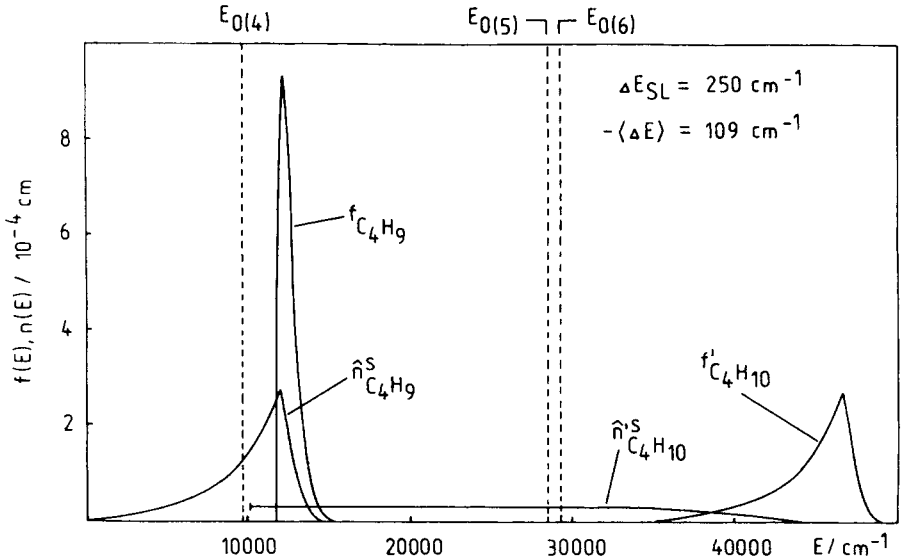


Figure 1. Normalized molecular distributions for step size ΔE_{SL} = 250 cm^{-1}. Symbols see text; $E_{0(4)}$, $E_{0(5)}$, and $E_{0(6)}$ see appendix. The lower break-off of the $\hat{n}^{'s}_{C_4H_{10}}$-curve is due to limited computer memory, but it has no effect on the remaining part of the curve.

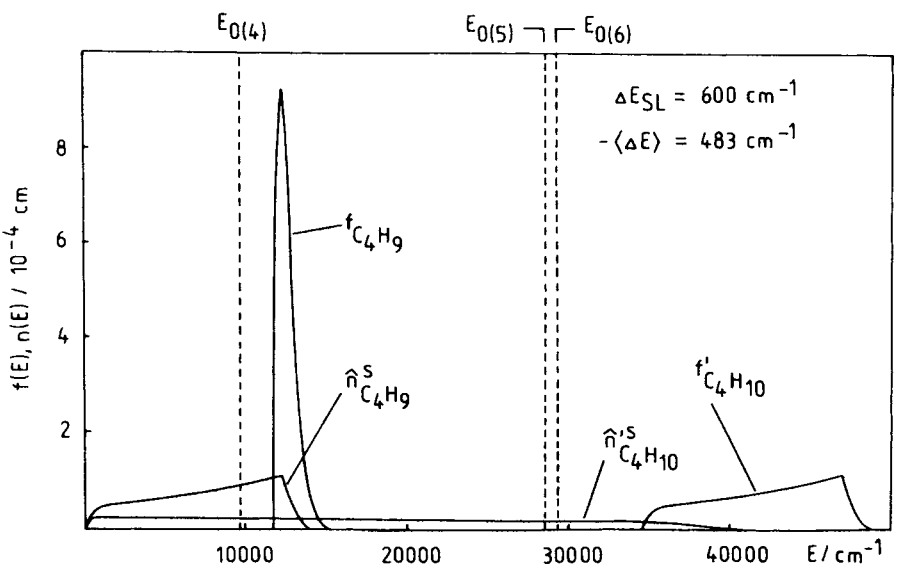


Figure 2. As Figure 1 but with step size ΔE_{SL} = 600 cm^{-1}.

Now, in analogy to eq. (2), a master equation for butane molecules is formulated:

$$R'\mathbf{F}' = (\omega\mathbf{I} - \omega\mathbf{P}' + \mathbf{K}')\mathbf{N}'^{s} = \mathbf{J}'\mathbf{N}'^{s}, \tag{9}$$

which leads to an expression similar to eq. (5):

$$\frac{D_2}{R'} = \sum_i (\mathbf{K}'\mathbf{J}'^{-1}\mathbf{F}')_i, \tag{10}$$

where $\mathbf{F}'$, $\mathbf{P}'$, $\mathbf{K}'$, and $\mathbf{N}'^{s}$ have the same meaning as above but refer to butane molecules (see Scheme II). Furthermore, the distinction between reactive and nonreactive collisions is unnecessary here.

Finally, with the steady-state condition for butane

$$R' = D_2 + S, \tag{11}$$

there are four equations, eq. (5), eq. (6), eq. (10), and eq. (11), with five unknown variables (D_1, D_2, R, R', and S), which allow the calculation of any ratio, especially D_2/D_1.

Calculations

In our calculations, we used RRKM theory for all microcanonical rate constants and a stepladder model, including upward transitions, for the collisional transition probabilities. All equations with a continuous energy scale, given in the text for notational simplicity, are substituted by their finite-difference analoga.

The input distribution for the 2-butyl radicals is given by [1,5]:

$$f(E) = \frac{G^{\ddagger}(E - E_0^{ar}) \exp(-E/k_BT)}{\int_{E_0^{ar}}^{\infty} G^{\ddagger}(E - E_0^{ar}) \exp(-E/k_BT)\, dE}, \tag{12}$$

where $G^{\ddagger}(E - E_0^{ar})$ is the integrated density of states for the transition state, and E_0^{ar} is the activation energy at absolute zero for the reaction

$$2\text{-}C_4H_9 \longrightarrow cis\ 2\text{-}C_4H_8 + H.$$

The collisional transition probabilities obey two constraints, completeness, eq. (13), and the principle of detailed balancing, eq. (14) [e.g., 4]:

$$P_{i+1,i} + P_{i-1,i} = 1 \tag{13}$$

and

$$\frac{P_{i+1,i}}{P_{i,i+1}} = \frac{\rho_{i+1}}{\rho_i} \exp\left(-\frac{E_{i+1} - E_i}{k_BT}\right), \tag{14}$$

where ρ_i denotes the density of states of the molecule at E_i. The same laws apply, of course, to the P'_{ij}. Equation (13) and eq. (14) represent two equations for three unknown variables, and the P_{ij} are not specified. Numerical tests with actual densities of states, however, show that in a very good approximation, eq. (13) can be substituted by

$$P_{i+1,i} + P_{i,i+1} = 1. \tag{15}$$

Using this relation, we obtain for a stepladder model with step size ΔE_{SL} the following transition probabilities:

$$P_{i+1,i} = \frac{A}{1 + A} \tag{16}$$

and

$$P_{i,i+1} = 1 - P_{i+1,i} \tag{17}$$

for $i = 1, \ldots, N - 1$, where N is the number of steps, and

$$A = \frac{\rho_{i+1}}{\rho_i} \exp\left(-\frac{\Delta E_{SL}}{k_B T}\right) \tag{18}$$

[see also 14]. With the exception of the first and the last step, all transitions are nearly exactly detailed balanced.

The input distribution of the butane molecules, $f'(E)$, follows from eq. (8). Because of the very narrow Maxwellian at room temperature, we use a δ-function for the distribution of the H-atoms, simplifying in this way the convolution integral, eq. (8), into a shift of the steady-state distribution of the 2-butyl radicals, $\hat{n}^s_{C_4H_9}$, by the bond dissociation energy D_0:

$$f'(E) = \hat{n}^s_{C_4H_9}(E - D_0) \tag{19}$$

with $D_0 = 34\,625\ \mathrm{cm}^{-1}$ [15].

Microcanonical unimolecular rate constants follow from RRKM theory [4,5]:

$$k(E) = L^{\ddagger}\frac{Q_1^{\ddagger} G^{\ddagger}(E - E_0)}{Q_1 F h \rho(E)} \tag{20}$$

with $L^{\ddagger}$ being the reaction path degeneracy, Q_1 the partition function for adiabatic rotations, F the centrifugal factor (Waage and Rabinovitch, see [4]), h the Planck's constant, and E_0 the activation energy at absolute zero for the reaction considered. $G^{\ddagger}$ and ρ are determined by direct count techniques [16], and ‡ refers, as usual, to transition state. For butane decomposition, we take the sum of the constants for both possible reactions.

The frequency of nonreactive collisions of 2-butyl radicals follows from the overall conventional gas kinetic collision frequency ω by:

$$\omega_{nr} = \omega - \frac{1}{4}\omega_{H/C_4H_9} \tag{21}$$

taken every fourth collision between H and 2-butyl to be reactive [17].

The solution of eq. (2) and eq. (9) to evaluate eq. (5) and eq. (10) was carried out not by direct inversion of the matrices $\mathbf{J}$ and $\mathbf{J}'$, respectively, but in a more convenient way by a simple Gaussian elimination algorithm for tridiagonal matrices [18].

The step size of the stepladder model, ΔE_{SL}, serves as the adjustable parameter. In a stepladder model which obeys detailed balancing, the connection between this step size and the average energy transferred in all

transitions (up and down), $\langle \Delta E \rangle$, is given by [8]:

$$\langle \Delta E \rangle = -\Delta E_{SL} \tanh\left(\frac{\Delta E_{SL}}{2F_E k_B T}\right), \tag{22}$$

with the factor F_E, which characterizes the density of states [19]:

$$F_E = \int_{E_0}^{\infty} \frac{\delta(E) \exp(-E/k_B T) dE}{\delta(E_\emptyset) \exp(-E_\emptyset/k_B T) k_B T}. \tag{23}$$

A summary of all numerical data can be found in the appendix.

Results and Discussions

Table I represents the dependence of the ratios of interest on the step size ΔE_{SL}, and Figures 1 and 2 show two examples for molecular distributions. At first, one can realize that the ratio D/S is much more sensitive to the step size than the ratio D_2/D_1. Our experimental finding, $D/S = 2.1$, is best represented by $\Delta E_{SL} = 570$ cm^{-1} which corresponds to $\langle \Delta E \rangle = -448$ cm^{-1}. In comparing these results with other works on chemical activation of the butyl system, one has to take care to relate to values obtained by equivalent methods. Besides the dependence of the model on the transition probabilities (exponential, stepladder), an especially interesting point is the question as to whether or not upward collisions are included in the data handling to extract an averaged ΔE from the experimental observable. In a system, containing upward and downward collisions, the average energies transferred are $\langle \Delta E \rangle_{up}$ and $\langle \Delta E \rangle_{down}$, respectively. If one neglects upward collisions, a smaller $\langle \Delta E \rangle_{down}$ results. If, on the other hand, upward collisions are included via detailed balancing $\langle \Delta E \rangle_{down}$ will become larger since it must compensate for the energy transferred in up collisions. The quantity, which actually remains the same, is the average energy transferred in all collisions (up and down), $\langle \Delta E \rangle$ and represents the net effect.

As a limiting case then, and provided the same model of transition probabilities, we would have to compare our $\langle \Delta E \rangle$, which was obtained under observation of detailed balancing, with $\langle \Delta E \rangle_{down}$ in cases where upward collisions are neglected *completely*. This problem, as well as the dependence on the transition-probability model, was extensively discussed by Snider [6, 8].

TABLE I. Influence of the average energy transferred per collision on the mechanism of decomposition, calculated with $F_E = 1.3$.

ΔE_{SL} / cm^{-1}	$-\langle \Delta E \rangle$ / cm^{-1}	D/S	D_2/D_1
180	58	240	1.5
250	109	23	2.1
300	152	9.9	2.4
400	252	4.1	2.9
500	365	2.5	3.4
570	448	2.1	3.6
600	483	1.9	3.7

For chemically activated butyl radicals in molecular hydrogen, Kohlmayer and Rabinovitch [20] found $\langle \Delta E \rangle_{down}$ to be about 870 cm^{-1} and 460 cm^{-1} for a stepladder and an exponential model, respectively. Tardy and Rabinovitch [21, see also 22] obtained a value of 450 cm^{-1} for a stepladder model. In their work, detailed balancing is accounted for in the $\mathbf{P}'$ matrix* that transforms the input distribution $\mathbf{f}^1$ into the steady-state distribution $\mathbf{N}^s$. However, in the equation for D/S, only down transitions are involved. Insofar as these different $\langle \Delta E \rangle_{down}$-values (including ours) are not directly comparable, nevertheless, they do not differ by a factor larger than 2, which on the other hand was found to be already inherent in the transition-probability models [20–22].

Thus, we have the fact that the energy transfer parameters which describe our extended mechanism of chemical activation agree in a satisfactory manner with the parameters determined from the classical mechanism, even though the former covers an energy range up to 50 000 cm^{-1}, and the latter reaches up to only 15 000 cm^{-1} excitation energy (see Figs.).

On the other hand, there is a large discrepancy to the $\langle \Delta E \rangle$-values determined in direct experiments, where amounts between -92 cm^{-1} and -175 cm^{-1} are given for collisions of H_2 with various excited hydrocarbon molecules [23,24]. With these values, our calculations lead to a D/S between 25 and about 7, respectively, which is much too large compared with our experiments. Despite of this effect on the ratio D/S, the influence on the reaction mechanism is rather limited as can be seen from Table I.

A variation of the energy transferred per collision does not lead to a qualitative change of the ratio D_2/D_1, which in each case is larger than one, indicating that the step via activated butane is more important than the step via the 2-butyl radical.

A very interesting point, beside the absolute magnitude, is the energy dependence of the average energy transferred per collision. In each case, we use the same value for the excited 2-butyl radicals (excitation energy about 13 000 cm^{-1}) and for the excited butane molecules (about 45 000 cm^{-1}). A possible change here would have a considerable influence on the ratio D_2/D_1.

This problem is still under discussion today [24,25], and as long as such influences are not to be taken into account exactly, there appears to be little reason in a more elaborated treatment of the collisional transition probabilities as well as of the microcanonical rate constants. The flat steady-state distribution of the chemically activated butane is influenced only to a very little extend by high pressures and/or large $\langle \Delta E \rangle$-values and in each case extends to energies above the threshold, in this way maintaining the reaction path, which contributes to D_2. This is equivalent to the problem of equipartition connected with the usual formulation of the steady-state high pressure rate constant for chemical activation [cf. 5; eq. (9.79) represents an averaging over equipartition]. Therefore, under steady-state conditions, it seems not possible to suppress the second way completely by more frequent and/or more efficient collisions but only by elimination of the chemical acti-

*Here, the symbols $\mathbf{P}'$, $\mathbf{f}^1$, and $\mathbf{N}^s$ are used in the sense of ref. [21]. They do not agree with the corresponding symbols used in this work.

vation via recombination, i.e., by a large excess of substrate. If it is impossible to achieve such an excess over the radical species, then both ways have to be considered.

Appendix

Temperature: $T = 298$ K. Collision parameters:

$$\sigma_{C_4H_9} \approx \sigma_{C_4H_{10}} = 0.54 \text{ nm} \quad [23],$$
$$\sigma_{H_2} = 0.283 \text{ nm} \quad [23]$$
$$\sigma_H = 0.19 \text{ nm} \quad [26],$$
$$\text{all } \Omega_{ik}^{(2,2)} = 1.$$

Frequencies in cm^{-1}, rounded, and moments of inertia in kgm^2, degeneracy in brackets: (1) 2-C_4H_9 [15]. 3100, 2960(8), 1440(8), 1300, 1200, 990(6), 960, 950, 400, 370, 350, $4.8 \cdot 10^{-47}$(2), $1.4 \cdot 10^{-46}$, internal rotations, free, $F_E = 1.30$; (2) n-C_4H_{10} [27, model IIIa]. 2965(4), 2925, 2920, 2885, 2880, 2860, 2855, 1470(2), 1460(2), 1455, 1450, 1380, 1375, 1360, 1300, 1285, 1260, 1180, 1150, 1055, 1010, 970, 950, 835, 805, 730, 430, 270, $5.1 \cdot 10^{-47}$(2), $1.88 \cdot 10^{-46}$, internal rotations, free, $F_E = 1.17$; (3) 2-$C_4H_9 \rightarrow$ *cis* 2-C_4H_8 + H. Transition state: 2960 cm^{-1} in 2-butyl is omitted (reaction coordinate), and the free rotation around the central C—C bond is substituted by 200 cm^{-1} [28], then all frequencies are multiplied by a common factor to give the high pressure Arrhenius data [15] ($T = 550$ K): $\lg(A_\infty/s^{-1}) = 12.62$ and $E_\infty = 145.2$ kJ/mol. 3275, 3130(5), 3125(2), 1525(3), 1520(5), 1375, 1270, 1050, 1045(5), 1015, 1005, 425, 390, 370, 210, $4.8 \cdot 10^{-47}$(2), internal rotations, free, $L^\ddagger = 2$, $Q_1^\ddagger/Q_1 = 1$, $E_0^{ar} = 11800$ cm^{-1} (to give the above E_∞); (4) 2-$C_4H_9 \rightarrow C_3H_6 + CH_3$. Transition state: 990 cm^{-1} in 2-butyl is omitted (r.c.), and free rotation around the central C—C bond is substituted by 150 cm^{-1} [28], then all frequencies are scaled to give at $T = 550$ K: $\lg(A_\infty/s^{-1}) = 12.97$ and $E_\infty = 122.7$ kJ/mol [15]. 3040, 2905, 2900(7), 1415(3), 1410(5), 1275, 1175, 970(5), 940, 930, 390, 360, 345, 145, $4.8 \cdot 10^{-47}$(2), internal rotations, free, $L^\ddagger = 1$, $Q_1^\ddagger/Q_1 = 1.8$, $E_{0(4)} = 9780$ cm^{-1} (corresponding to E_∞). The ratio $Q_1^\ddagger/Q_1$ is calculated with $(I_A I_B I_C)^\ddagger = 3.79 \cdot 10^{-135}$ kg^3m^6 [estimated from ref. 29] and $I_A I_B I_C = 1.22 \cdot 10^{-135}$ kg^3m^6 [15]; (5) n-$C_4H_{10} \rightarrow 2$ C_2H_5 [27, model III(a)]. Transition state: all frequencies and moments of inertia are the same as in n-C_4H_{10} with the following exceptions: 835 → r.c., 1150 → 245, 970 → 200, 430 → 80, 270 → 45, $L^\ddagger = 1$, $Q_1^\ddagger/Q_1 = 2$, $E_{0(5)} = 28485$ cm^{-1}. The corresponding microcanonical rate constants are given in Table II; (6) n-$C_4H_{10} \rightarrow C_3H_7 + CH_3$ [27, model III(a)]. Transition state: all frequencies and moments of inertia are the same as in n-C_4H_{10} with the following exceptions: 1010 → r.c., 1150 → 220, 970 → 180, 430 →

TABLE II. Microcanonical rate constants for ethyl rupture from butane.

	$k(41895\ cm^{-1})/s^{-1}$	$k(43120\ cm^{-1})/s^{-1}$
Experiment [27]	$4.9 \cdot 10^6$	$1.5 \cdot 10^7$
RRKM [this work]	$7.1 \cdot 10^6$	$1.4 \cdot 10^7$

TABLE III. Microcanonical rate constants for methyl rupture from butane.

	$k(41895\ cm^{-1})/s^{-1}$	$k(43120\ cm^{-1})/s^{-1}$
Experiment [27]	$2.4 \cdot 10^6$	$7.7 \cdot 10^6$
RRKM [this work]	$2.9 \cdot 10^6$	$6.2 \cdot 10^6$

$80, 270 \rightarrow 50, 1.88 \cdot 10^{-46} \rightarrow 120, 5.1 \cdot 10^{-47} \rightarrow 2.85 \cdot 10^{-46}$, $L^{\ddagger} = 2$, $Q_1^{\ddagger}/Q_1 = 2$, $E_{0(6)} = 29300\ cm^{-1}$. Microcanonical rate constants see Table III.

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